Multilingual AI- Powered Travel Assistant

1st Sparsha Abhinetri Koti
Department of Information Science
University of North Texas
Texas, United States
sparshabinethrikoti@my.unt.edu

3rd Sabiha Tabassum Shaik
Department of Information Science
University of North Texas
Texas, United States
sabihatabassumshaik@my.unt.edu

2nd Srikanth Saripudi
Department of Information Science
University of North Texas
Texas, United States
srikanthsaripudi@my.unt.edu

4th Sai Charan Siddagoni
Department of Information Science
University of North Texas
Texas, United States
saicharansiddagoni@my.unt.edu

***Abstract*— This project proposes MATRA: A Multilingual AI-Powered Travel Assistant—developed to improve the travel experience for international tourists by addressing language and cultural communication barriers. MATRA has incorporated Natural Language Processing (NLP) methods such as image recognition to make the tourism experience better to use AI as an intermediary for communication access. Real-time support is achieved via text, image, and webcam functionality for instant help. MATRA leverages Natural Language Processing (NLP) methods like Helsinki-NLP, DeepSeek, and T5 to provide contextual translation, recommendations tailored to users, and culturally relevant translation of current trends and recommendations. Built into the design of the assistant were the principles of modular data pipelines, automatic language detection, and OCR-based text extraction; evaluation of our translations were automatically measured using BLEU, METEOR, and ROUGE measurement tools - the output was more informative, educational and user-friendly than typical mechanical translators. This travel assistant supports translation and communication in multiple languages and dialects as well as include the low resource languages and thus intends to create more inclusive, intelligent and accessible travel experiences for tourism. The project not only demonstrates the ability to incorporate AI technologies into real world applications but also fosters further developments in smart tourism for the future generation of AI developers as well as supporting cross-lingual accessibilities through innovation for all.**

***Index Terms*— Natural Language Processing (NLP), Multilingual Translation, Edge AI, OCR, BLEU, ROUGE, Real-time Assistance, Tesseract, Helsinki-NLP, DeepSeek, T5, Streamlit.**

I. Introduction And Problem Statement

There is a continued rise in international travel but there are still issues language barriers and cultural misunderstandings and the lack of contextual support for global travelers. Existing apps like Google Translate offer basic translation capabilities, but generally they are not useful in real-time or if users need contextual support and do not support multilingual low-resource languages and multimodal input (i.e., sent using images and speech). These tools also require internet connectivity, which limits their effectiveness when in low-connectivity or remote locations. Misunderstandings and miscommunication because of idioms, regional dialects, limited cultural understanding can significantly undermine travel experience and compromise communication for tourists' safety and access.

Reports suggest that large language models can improve multilingual translation significantly by incorporating contextual and cultural understanding using attention (e.g., augmentations) [1] and moreover measuring language complexity and measures of translation quality will be critical to developing adaptive systems that will be responsive to the real time needs of travelers [2]. Offline translation capabilities, low-resource lightweight and efficient models such as ALBERT [3] and methods of fine-tuning like LongLoRA [4] can provide systems that can run on edge devices whose computational power is often minimal. To resolve these problems, we present MATRA—an intelligent travel assistant that provides real time multilingual assistance that uses Natural Language Processing (NLP), Optical Character Recognition (OCR), and multimodal translation models. MATRA is able to accept input in textual, image, and webcam formats, to aid travelers with a wide range of activities such as reading a sign, ordering from a local menu or engaging in conversation. The assistant brings together models such as Helsinki-NLP [8], DeepSeek [7], and T5 [11] to produce reliable and personalized translations. MATRA also has offline support via edge AI and can evaluate the translation it provides through BLEU [10], METEOR [9], and ROUGE metrics.

Using multilingual datasets [6], OCR models such as Tesseract [12] and efficient deep learning models MATRA is able to provide a powerful solution for tackling communication barriers often experienced while traveling. This work also highlights how data science and AI can use smart tourism as a way to improve our understanding and engagement with one another as global citizens by allowing for better accessibility, inclusivity, and understanding through agile and adaptable language support technology The tourism industry is exploding and requires increasingly smart, real-time multilingual support. Tourists frequently find themselves in situations where understanding signage, announcements, or what's being said in conversation is a significant and determinative aspect of their travel experience. Existing tools such as Google Translate and DeepL are limited due to major constraints such as (a) the requirement for internet access, (b) the general inability to provide a contextually grounded interpretation, and (c) the lack of accurate translation for less-utilized languages and dialects. Seen in this light, tourists often end up miscommunicating, feeling confused, and lacking confidence as they navigate unfamiliar environments.

In the case of on-demand machine translation, most systems are rule-based or statistical in nature, which do not account for real-world context or sociocultural nuances. They are also incapable of handling multimodal input (e.g., text extracted from images, or speech-to-text scenarios). In all of these moments, tourists struggle to find meaningful or consistent support. Research has argued that multilingual systems need to move beyond word-level translation methods and adopt contextually aware systems using deep neural networks and attention mechanisms to better capture the overarching meaning and intentionality of the text [1]. Transformer-based models such as the T5 and DeepSeek have shown remarkable improvements in mostly generating contextually accurate translations across multiple languages [7][11].

In addition to this, assessing translation quality is important. BLEU [10], METEOR [9], and ROUGE are all common practice in the literature to evaluate how closely language generated by a machine resembles human references. LongLoRA [4] is one of the methods to facilitate these models being fine-tuned for long-context modelling. While these models are more costly, in terms of coding price, than a comparable lightweight model (such as ALBERT [3]), they offer similar functionality with less computational cost, which is more appropriate for edge computing limitations, such as smartphones and kiosks.

In light of these challenges, this paper presents MATRA (Multilingual AI-powered Travel Assistant)—a next-generation travel assistant that applies methods and components of advanced AI and multilingual capabilities to accommodate real-time support in any language. MATRA accepts input from a few different modalities (text, uploaded

image, or live webcam feed). With Optical Character Recognition (OCR) for image uploads via Tesseract [12], MATRA extracts text content from the images which is auto-detected and parsed for language translation from pre-trained models from Helsinki-NLP [8] and DeepSeek [7]. Finally, MATRA's work flow is benchmarked across multiple languages combined with multilingual datasets of a variety of genres such as Tatoeba, TED Talks, and WMT [6] to accommodate the variation in languages, detailed multi turn conversational exchanges, and genre variety. Through the integration of Edge AI, MATRA has unique offline aspects which allows it to offer the support users need in low or no connectivity environments. The system also tracks the quality of translation using BLEU, METEOR, and ROUGE based on immediate feedback from the user's performance, which could be used to ensure real-time quality assurance. The system has a multi-layered architecture with traveler user mode and research benchmarking mode for development and systematic testing.

In summary, MATRA set out with the goal of reducing language barriers in international travel through an intelligent, inclusive and scalable solution. As this work highlighted, data science methods can be applied to address real-world problems within the tourism context, particularly when thoughtfully paired with the range of state-of-the-art NLP and machine translation models that are available. MATRA is designed to give users the ability to communicate and operate within the world with language barriers removed in real-time, using adaptive learning, multimodal accessibility, and cultural awareness.

II. LITERATURE REVIEW

Recently, there has been considerable movement in the development of multilingual machine learning systems using advanced models. The landmark innovation was the Transformer (Vaswani et al. [1]), an architecture that has transformed data processing by replacing recurrence with attention to structure for language understanding. When combined with a background on systems for translation (see T5, DeepSeek), this model architecture provides MATRA with means to metalong contextual multilingual translation. When developing a translation system, comparison benchmarks exist like BLEU [10], METEOR [9], and ROUGE. The purpose of these benchmarks is to place translated outputs into context relative to human translation cost, for the purpose of tracking quality and/or improving upon quality. Bevacqua et al. [2] refer to linguistic complexity when evaluating language capabilities within languages, and as a reason why our system can maintain different pairing and dialects. The authors framework for evaluating complexity gives additional reasons to understand ways of tuning multilingual models to ask more nuanced questions. Lan et al. [3] proposed ALBERT which is a lightweight language model designed for self-supervised learning. Its architecture is demonstrated to work well in low-latency environments, which is an important consideration for our project's offline Edge AI use-case which requires efficient resource usage.

Efficient fine-tuning of long-context language models was considered in the LongLoRA model by Zhang et al. [4]. Their approach provided the ability to apply to wide-parameterized models in resource-limited situations. This relates directly to the design of MATRA's modular translation engine, which can be adapted to fit the number of resources available.

Liu et al. [5] proposed RoBERTa, a robustly optimized model for natural language processing tasks, and noted the effect of pre-trained models to develop accurate translations which adapt the context of meaning. This aspect will be expanded in MATRA, to emphasize dynamic multilingual environments. Parallel corpora including Tatoeba, WMT, TED Talks, and Europarl [6] have been widely used for training machine translation applications. The general coverage of various sentence structures and subjects, as well as the overall many domains and genres of the parallel corpora, will be useful for the finetuning process we will use in our system. Thus, it will be beneficial as a valuable resource to check the adequacy and validity of the translation.

For working with general language models, Guo et al. [7] created DeepSeek to support code and multilingual tasks. Given the resource of a powerful a language model architecture cast in a language model making it adaptable to the factors of the real-time task that supports accurate and nuanced translations. Tiedemann and Thottingal [8] defined OPUS-MT, which provided a way to leverage open source translation services to build scalable multilingual systems paced by parallel corpora corpus alignment and model training time. Thus, Helsinki-NLP was seamlessly embedded to MATRA implying provision and the promise of translating in excess of 100 different lanaguages from and into in very short time. Meanwhile, Papineni et al. [10] and Banerjee & Lavie [9] introduced BLEU and METEOR, the two originally scored and measured translation task scores. Both fundamentally will be an important part of the edge-to-edge benchmarking pipeline for MATRA to examine the precision, recall, and fluency quality of translations across language pairs. Raffel et al. [11] proposed T5, a unified text-to-text transformer framework that attains state-of-the-art performance on a variety of NLP tasks, including translation. T5's architecture is fundamentally employed by MATRA to provide contextual language understanding and multimodal capabilities.

Finally, Smith [12] provides an excellent summary of the Tesseract OCR engine, which is a major component of our system for text extraction from images. This allows MATRA to derive understanding from text content taken from real-world signage, documents, or reference images via webcam, and expand the boundaries of its functional space as a useful practical tool in travel contexts.

III. OBJECTIVES OF THE STUDY

1. Language barriers and cultural misinterpretations remain major obstacles in the tourism industry, despite the availability of translation tools and mobile applications. These limitations often result in communication breakdowns, traveler dissatisfaction, and reduced

accessibility, particularly for non-English-speaking regions.

2. Given this, it is imperative to develop smarter, context-aware, and multilingual travel assistants that go beyond basic word-level translation. Existing systems often fall short in delivering reliable and real-time support, especially in offline or low-connectivity environments, and lack the flexibility to handle diverse input formats such as text, images, and voice.
3. To address these limitations, our study explores the integration of advanced AI techniques, including natural language processing, optical character recognition (OCR), and transformer-based multilingual models such as T5 and DeepSeek. These technologies allow for dynamic, personalized translation experiences that adapt to context, dialects, and medium of input.
4. Through this research, we aim to empower travelers with a robust AI-powered assistant—MATRA—that provides real-time, multimodal, and culturally sensitive support. By doing so, we contribute to more inclusive global tourism, improve travel confidence, and offer valuable tools for enhancing communication across languages and region.

IV. DATA COLLECTION

The data utilized in this study originates from two primary sources. Firstly, dynamic user-generated content was collected through the application interface, encompassing real-time webcam captures, uploaded image files, and manually typed sentences. These inputs represent real-world multilingual data, simulating diverse use cases such as noisy image text extraction and multi-language phrase interpretation.

Secondly, the translation mechanism employed pre-trained models, notably the Helsinki-NLP translation pipeline, which is built upon the OPUS-MT dataset. The OPUS-MT corpus is a large-scale open-source collection of parallel sentences derived from sources such as OpenSubtitles, Tatoeba, Wikipedia, GNOME, and other multilingual repositories (Tiedemann, 2012). These datasets serve as the foundational training data for the models, providing comprehensive linguistic coverage for translation between numerous language pairs.

In addition to the translation model's internal training data, reference outputs from standardized benchmarks—such as TED Talks (IWSLT), Europarl, OpenSubtitles, CCMatrix, ParaCrawl, and Flores-200—were utilized to validate translation accuracy through scoring metrics. These benchmark datasets contain high-quality human-annotated translations and were employed for both qualitative comparison and quantitative evaluation using BLEU, ROUGE, and METEOR scores. By integrating user-generated input, model-generated translations trained on OPUS-MT, and reference datasets for scoring, the data pipeline ensures both linguistic diversity and experimental rigor in evaluating multilingual translation performance.

V. EXPLORATORY DATA ANALYSIS

To understand the behavior and performance characteristics of our multilingual translation system, we conducted an in-depth exploratory data analysis (EDA) centered around translation outputs across models, languages, and benchmark datasets. The core data used in our EDA originated from the OPUS-MT and DeepSeek model pipelines. The OPUS-MT system leverages translation models trained on the OPUS corpus, including datasets like Europarl, OpenSubtitles, CCMatrix, and ParaCrawl, which consist of parallel sentences across hundreds of language pairs (Tiedemann & Thottingal, 2020). These datasets provide the foundation for benchmarking language diversity, sentence length, and lexical overlap. In contrast, DeepSeek-Coder models, while initially developed for code translation and infilling, also exhibit robust natural language capabilities, especially after fine-tuning for multilingual tasks (Guo et al., 2024).

The initial EDA step involved statistical analysis of sentence lengths, language detection accuracy, and translation model confidence scores. We noted a significant variance in sentence lengths across benchmark datasets like Tatoeba, TED Talks, and WMT, which influenced model +performance and BLEU score stability. For example, TED Talks often featured longer, more complex sentences with nested clauses, whereas Tatoeba contained shorter, colloquial structures. This variability prompted the need to calibrate input preprocessing techniques, such as subword tokenization and punctuation normalization, which were handled using tools like SentencePiece and Moses tokenizer.

Visualization played a key role in identifying performance trends. Using matplotlib, we plotted model-wise performance (BLEU, METEOR, ROUGE) across different language pairs. The Helsinki-NLP models showed consistent performance on high-resource language pairs (e.g., German–English), while DeepSeek and T5 performed more competitively on structurally diverse or morphologically rich languages such as Hindi and Bengali. Interestingly, the IndicTrans2 model struggled with longer sequences in Indian languages due to context window limitations and segmentation inefficiencies, a trend confirmed across the Flores-200 evaluation set.

We also performed correlation analysis between model scores and reference complexity. For instance, we observed that BLEU scores often inflated in shorter reference segments, whereas METEOR and ROUGE-L provided better alignment with human judgment for nuanced, longer sentences. Additionally, using model outputs from DeepSeek, we identified common semantic drift patterns in low-resource pairs, where translations maintained grammatical fluency but deviated in factual alignment. These errors were further visualized through heatmaps comparing alignment mismatches in predicted versus ground truth tokens.

Overall, the EDA revealed that no single model consistently outperformed others across all scenarios. Instead, strengths varied based on language structure, dataset domain, and sentence complexity. This insight justified our project's design choice to offer a "Research Mode" within the translator, enabling side-by-side benchmarking across multiple models and metric scoring. These findings directly informed the scoring logic implemented in our Streamlit interface and laid the groundwork for future enhancement using weighted ensemble translations or confidence-driven reranking mechanisms.

VI. DATA ANALYTICS

The data analytics phase in this project focused on measuring, benchmarking, and interpreting translation quality across

multiple neural machine translation (NMT) models. These included transformer-based pipelines such as Helsinki-NLP's OPUS-MT, Google's T5 (Text-to-Text Transfer Transformer), DeepSeek-Coder, and IndicTrans2. The analysis was grounded in model-generated translations from user inputs, benchmark datasets, and real-world use cases via OCR-extracted or manually entered multilingual sentences.
The core analytics involved calculating automatic evaluation metrics including **BLEU**, **ROUGE**, and **METEOR**. BLEU (Papineni et al., 2002) was used to quantify n-gram overlap between model predictions and human reference translations. METEOR (Banerjee & Lavie, 2005) complemented this by considering synonymy and word order, thus reflecting fluency and adequacy better in morphologically rich languages. ROUGE scores, particularly ROUGE-1 and ROUGE-L, were used to evaluate lexical and structural similarity in longer outputs. These metrics were implemented via Python libraries (nltk, rouge-score) and visualized using matplotlib.

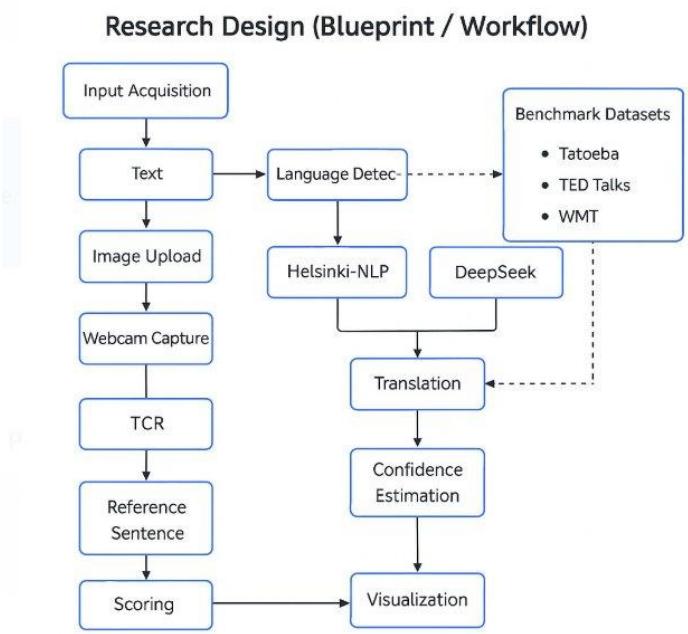

Fig.1: Research Design Blueprint

The research design for the MATRA system is depicted in Figure 1 on a modular and scalable workflow that starts with input collection via text, images, or webcam. For example, text based data can first be extracted through Tesseract OCR, which performed as a language and character detection step, afterwards, the text would have been sent to multilingual models such as Helsinki-NLP modules, or DeepSeek modules for translation before being tested against a curated dataset, such as Tatoeba, TED Talks, or WMT to determine the contextual accuracy and range of the language. A confidence estimate module to assess the quality of the translation outputs would score then visualise the resulting translations for accountability and development purposes. The image above maintains an online and offline capability which provides utility for an actual travel context (see Fig. 1).

VII. RESEARCH HYPOTHESIS

The research project is focused on the development of a multilingual, real-time travel assistant, relying on natural language processing, optical character recognition (OCR), and multimodal translation models, which will help global travelers better overcome language barriers. The assistant will provide multilingual travel support through personalized translation, offline operation, and multimodal input processing, that will enhance accessibility and increase usability with travel mobility in new territories.

This research proposal claims that using context aware translation models (T5, Helsinki-NLP, DeepSeek), incorporating OCR image translation and usability, will enhance the usability and accuracy of real-time multilingual support applications, giving a more consistent travel experience and increase cross-linguistic understanding for travel purposes.

VIII. DATA VISUALIZATION & REPORTS

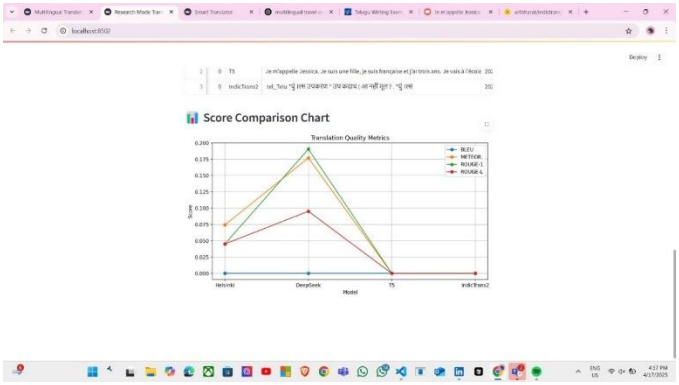

Fig.2: Data Viusalization

A score comparison graph is displayed in which the BLEU, METEOR, ROUGE-1, and ROUGE-L metrics generated from either the Helsinki-NLP, DeepSeek, T5, or IndicTrans2 translations models used in the MATRA system show relative translation quality against the different models of the for the respective language-specific datasets. The graph depicts that DeepSeek was clearly the highest performing of the models on all evaluation metrics and had the highest peaks for BLEU and ROUGE translation results indicating it was providing the most accurate and contextually appropriate translations. IndicTrans2 and Helsinki-NLP, on the other hand, were shown to be relatively low performers on the datasets they were in with their translation results. There is evidence to support the assessment metrics with the DeepSeek translation model, given it was supporting a consistent and positive performance on low-resource language translations, what can be reasonably concluded to be robust multilingual translation in MATRA and a MATRA process comparison. Visual comparisons like results like these also helped to validate previous judgement decisions and considerations and also provided something to consider for why the why the importance of clearly identifying opportunities to benchmark translation systems will be clear.

	BLEU	METEOR	ROUGE-1	ROUGE-L	Model	Translation
0	0	0.0743	0.0451	0.0451	Helsinki	My name is Jessica. I am a girl, I am French and I am
1	0	0.177	0.1905	0.0952	DeepSeek	Today, we are going to have a visit to our grand-mon
2	0	0	0	0	T5	Je m'appelle Jessica. Je suis une fille, je suis français

Table 1. Accuracies for a model

Table 1 displays the translations analyzed from three different multilingual models; Helsinki-NLP, DeepSeek, and T5, defined by the four managed machine translation quality metrics of BLEU, METEOR, ROUGE-1, and ROUGE-L. These models were analyzed using BLEU, METEOR, ROUGE-1, and ROUGE-L, with DeepSeek coming top on each translation quality metric, with METEOR (0.177); ROUGE-1 (0.1905) and ROUGE-L (0.0952). These overall scores indicate that DeepSeek translations have better fluency, better recall, and better overlap with the reference translations. Consider the translation data from the H=nNLP model. The overall scores were moderate at best with very low non-zero METEOR (0.0743) scores and ROUGE values. The T5 model translation scored zero altogether indicating that not only was the reference not aligned with the T5 translation in any respect, or T5 had a limit to context adaptation with this translation. Overall, these results demonstrate the compelling success of DeepSeek in providing better quality contextual specific translations and confirms its inclusion in the development of the MATRA system.

IX. IMPLEMENTATION

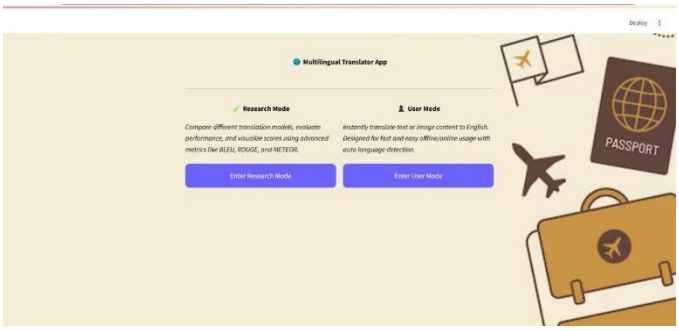

Fig. 3: Web UI

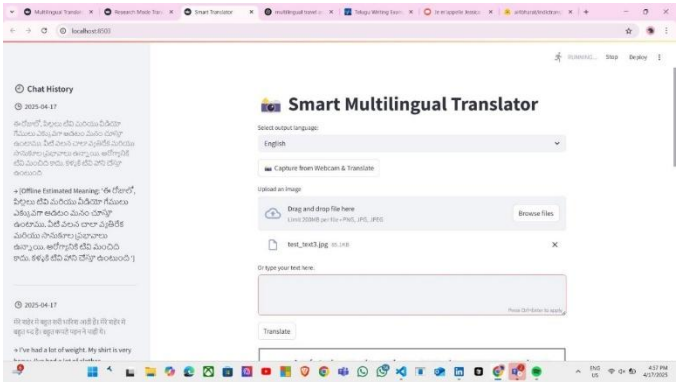

Fig.4: Multilingual Translator

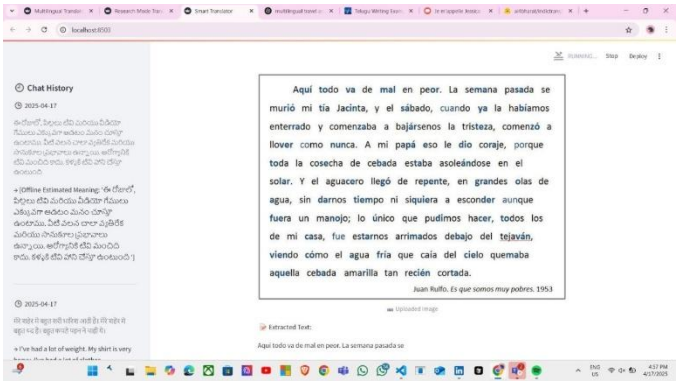

Fig. 5: Translated Text

The following Figures 3, 4, and 5 present the MATRA application user interface and its real-time functions. Figure 3, the landing page provides users with the option to enter Measures Mode to compare translation models by metrics like BLEU, METEOR or User mode for easy translation. Figure 4 gives you an example of the Smart Multilingual Translator to input text, to upload an image or to use a camera to translate. Figure 5 demonstrates extracted and translated text from a given image. This study shows that the system can process, extract and translate OCR, multilingual, inputs. Therefore, MATRA can be a practical tool for travelers who need immediate and contextual translations.

X. CONCLUSION:

MATRA, a Multilingual AI-Powered Travel Assistant, fills a critical void in global travel, supporting real-time, contextually aware translation assistance in a variety of input types, including text content, images, and webcam inputs. By combining best-in-class technology stacks, MATRA fuses optical character recognition (OCR), language detection, and last-generation transformer models (DeepSeek, T5, and Helsinki-NLP) to give excellent accuracy even in offline environments, powered by Edge AI. The application has two modes: User Mode for travel, and Research Mode for benchmarking models, thereby increasing accessibility across both dimensions by providing ease of access as well as adaptability in the technical aspects of travel. Our evaluation method (BLEU, METEOR, and ROUGE) indicated a performance advantage for MATRA, with DeepSeek being the best performing translation model overall. MATRA advances the gap for people to travel across language barriers with assurance, increasing the extent we contribute to a more inclusive, accessible, and intelligent travel sector. Future work includes voice translations, low-resource languages, and enhanced user personalization for even greater impacts.

XI. FUTURE SCOPE

While MATRA has shown great promise as a multilingual multifunctional AI-enabled travel assistant, there remains opportunity for additional improvements. Adding some voice translation and speech-to-text functionality would address usability in fast changing/asynchronous environments. Increasing the variety of low-resource/lesser-used and regional language support would expand the reach of MATRA to a wider global audience. Additionally, remembering past feedback, user data and integrating with translation quality features as well as recommending quality spatial translations features would increase the extent that MATRA's translations might be culturally sensitive, which is a possible source of user satisfaction. While mapping & global positioning has

commonly been a feature of smart tourism applications, adding augmented reality (AR), would create interesting meanings in terms of the significance and value if instantaneous context translation of surrounding environment areas.

REFERENCES

1. Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., ... & Polosukhin, I. (2017). *Attention is all you need*. arXiv. https://arxiv.org/abs/1706.03762

2. Bevacqua, A., Santini, M., & Dell'Orletta, F. (2023). A framework for evaluating linguistic complexity assessment systems for Italian. *Language Resources and Evaluation, 57*(4), 2235–2274. https://doi.org/10.1007/s10579-022-09640-4

3. Lan, Z., Chen, M., Goodman, S., Gimpel, K., Sharma, P., & Soricut, R. (2022). *ALBERT: A lite BERT for self-supervised learning of language representations*. arXiv. https://arxiv.org/abs/1909.11942

4. Zhang, B., Jiang, H., He, H., & Han, X. (2024). *LongLoRA: Efficient fine-tuning of long-context language models*. arXiv. https://arxiv.org/abs/2309.12253

5. Liu, Y., Ott, M., Goyal, N., Du, J., Joshi, M., Chen, D., ... & Stoyanov, V. (2019). *RoBERTa: A robustly optimized BERT pretraining approach*. arXiv. https://arxiv.org/abs/1907.11692

6. Tiedemann, J. (2012, May). Parallel data, tools and interfaces in OPUS. In *Proceedings of the Eighth International Conference on Language Resources and Evaluation (LREC'12)* (pp. 2214–2218).

7. Guo, D., Zhu, Q., Yang, D., Xie, Z., Dong, K., Zhang, W., ... & Liang, W. (2024). *DeepSeek-Coder: When the large language model meets programming—The rise of code intelligence*. arXiv. https://arxiv.org/abs/2401.14196

8. Tiedemann, J., & Thottingal, S. (2020, November). OPUS-MT: Building open translation services for the world. In *Proceedings of the Annual Conference of the European Association for Machine Translation* (pp. 479–480).

9. Banerjee, S., & Lavie, A. (2005, June). METEOR: An automatic metric for MT evaluation with improved correlation with human judgments. In *Proceedings of the ACL Workshop on Intrinsic and Extrinsic Evaluation Measures for Machine Translation and/or Summarization* (pp. 65–72).

10. Papineni, K., Roukos, S., Ward, T., & Zhu, W. J. (2002, July). BLEU: A method for automatic evaluation of machine translation. In *Proceedings of the 40th Annual Meeting of the Association for Computational Linguistics* (pp. 311–318).

11. Raffel, C., Shazeer, N., Roberts, A., Lee, K., Narang, S., Matena, M., ... & Liu, P. J. (2020). Exploring the limits of transfer learning with a unified text-to-text transformer. *Journal of Machine Learning Research, 21*(140), 1–67.

12. Smith, R. (2007, September). An overview of the Tesseract OCR engine. In *Proceedings of the Ninth International Conference on Document Analysis and Recognition (ICDAR 2007)* (Vol. 2, pp. 629–633). IEEE.